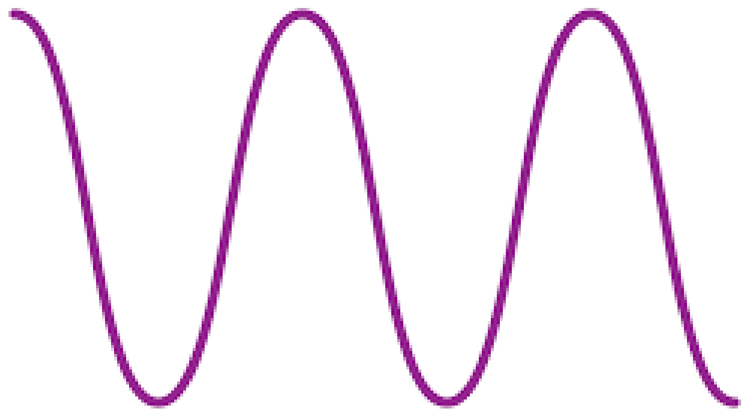
Waves and Oscillations

OSCILLATORY MOTION

Oscillatory motion is defined as the to and fro motion of the body about its fixed position.

Waves

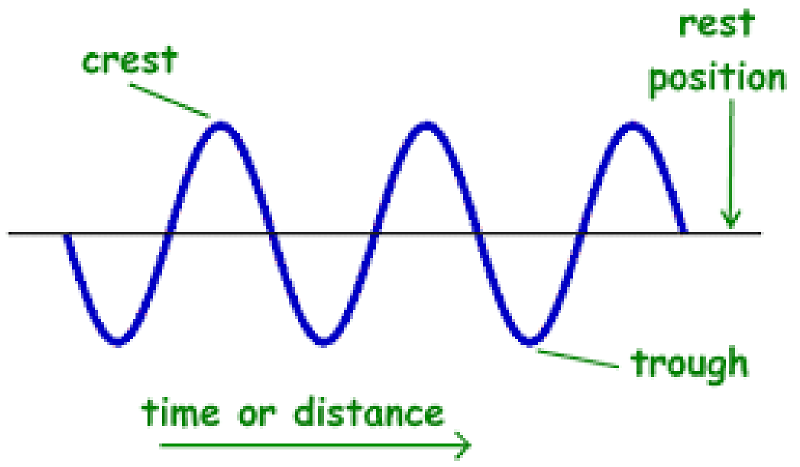
- Carriers of energy that is used to transmit energy from one place to another in a medium
- It can be described as a disturbance or pattern that travels through a medium from one location to another location.



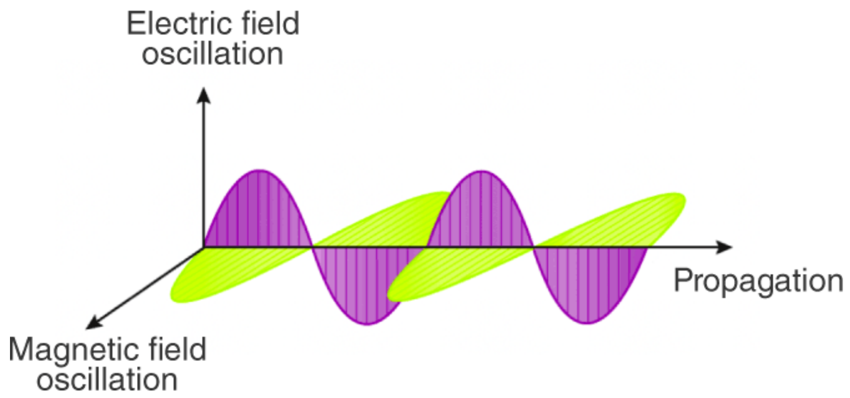
Types of Wave

With respect to Medium of Propagation

- **Mechanical Wave**
  - Requires medium to go from one place to another
  - Examples, water waves and sound waves



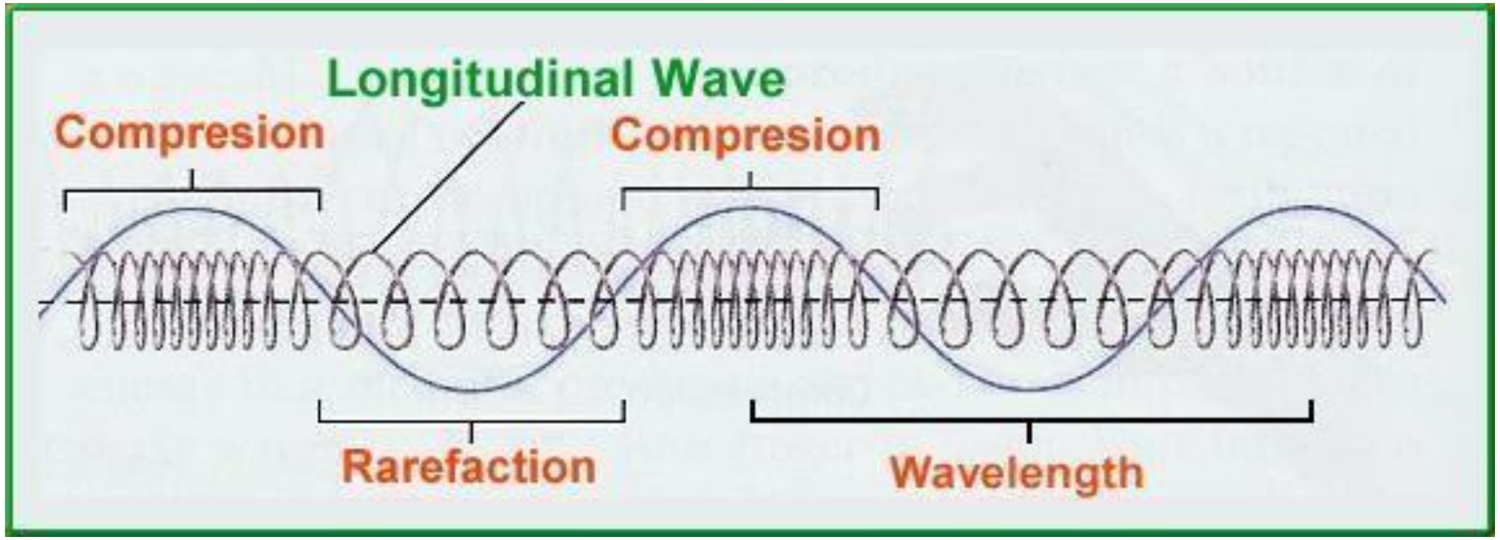
- **Electromagnetic Wave**
  - Do not require medium to go from one place to another
  - Examples, TV rays, X-rays and Radar waves



With respect to Direction of Propagation

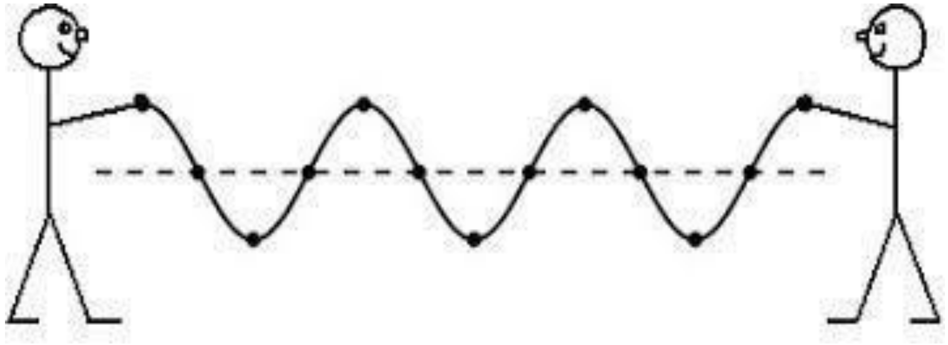
Longitudinal waves

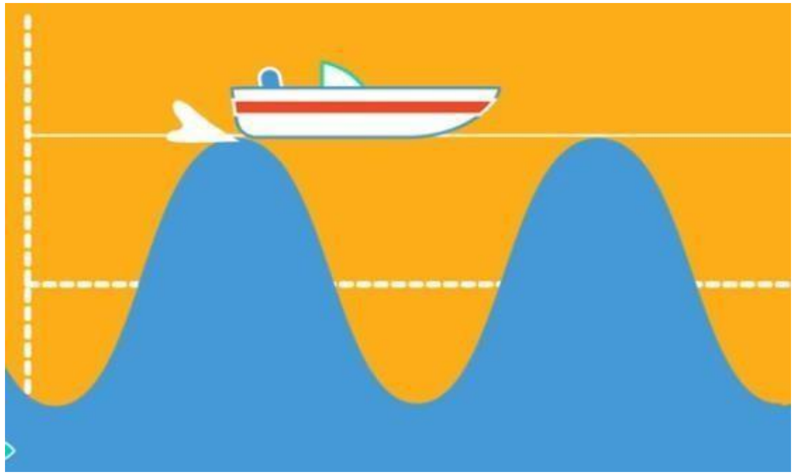
- Waves in which the direction of wave motion is parallel to the motion of particle in a medium For example, Sound waves



Transverse waves

- Waves in which the direction of wave motion is perpendicular to the motion of particle in a medium
- For example, Water waves

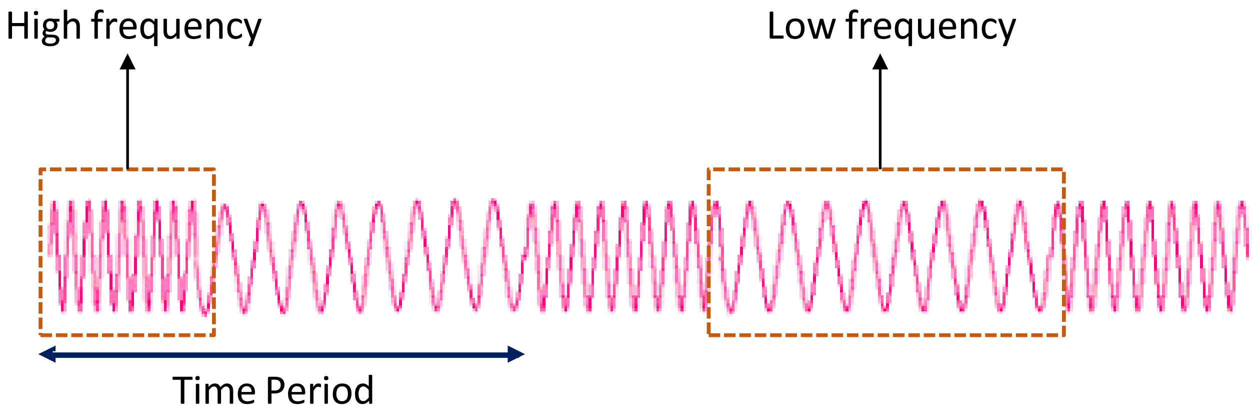




Frequency and Time Period

- **Frequency**
  - The no of oscillations per second
  - Measured in “hertz” or “Hz”
  - 1 Hz = 1 oscillation per second

- **Time Period**
  - Time required for one complete oscillation or cycle
  - Measured in “second” or “sec”
  - T = $\frac{1}{f}$ = $\frac{1}{Hz} = sec$

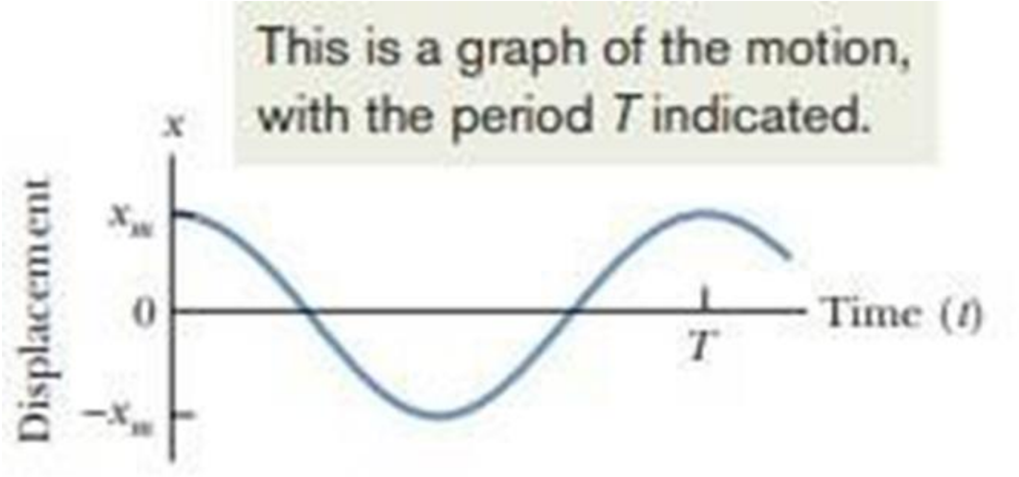


Simple Harmonic Motion

The kind of motion of a body in which acceleration is directly proportional to displacement x(t)

$$x(t) = x_m \cos(\omega t + \phi) \quad \text{(displacement)},$$

- $\cos(wt + \phi)$ = Phase
- w = Angular frequency = $2\pi f = \frac{2\pi}{T}$ (rad/sec)
- ϕ = Phase angle / Phase constant

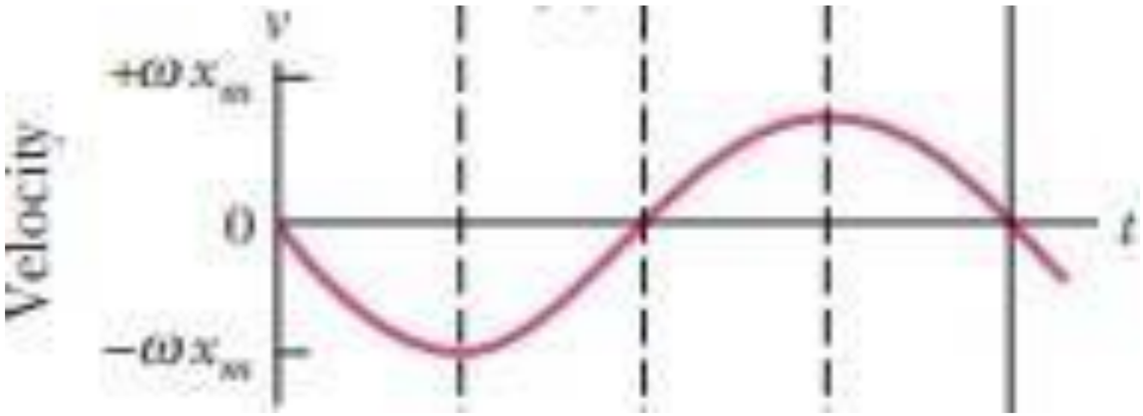


Simple Harmonic Motion

x(t) = xmcos(wt + ϕ)

$\frac{dx}{dt}$ = - xm sin(wt + ϕ) x (w)

v(t) = - w xm sin(wt + ϕ) → Velocity

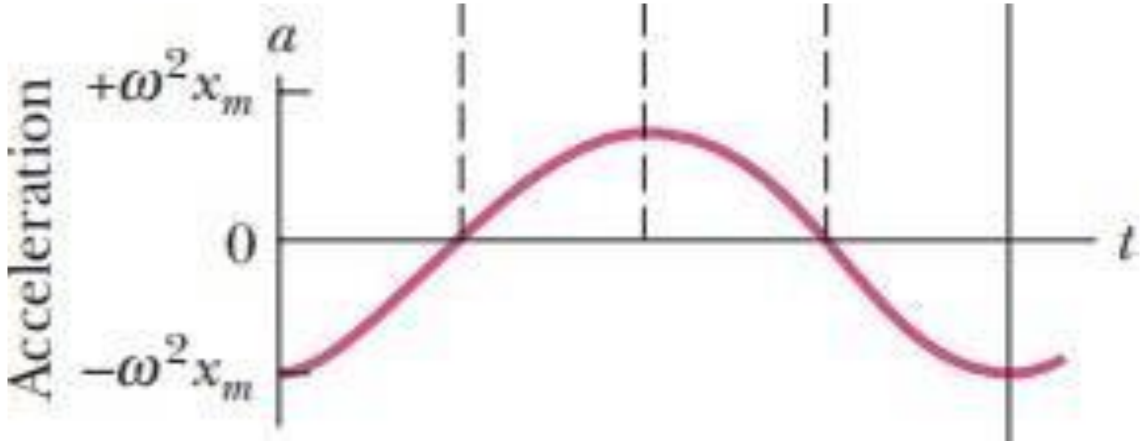


Simple Harmonic Motion

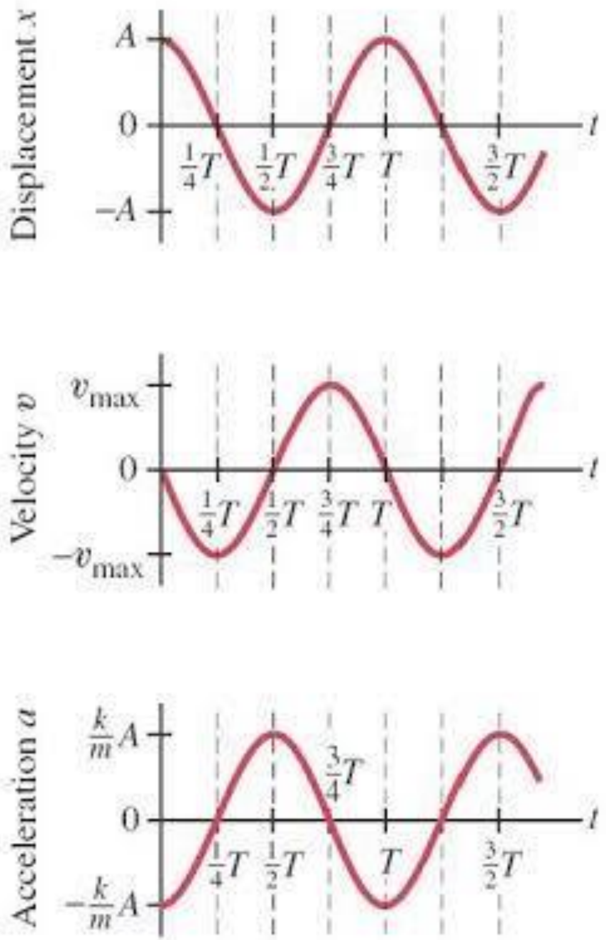
v(t) = - w xm sin(wt + ϕ)

$\frac{dv}{dt}$ = - w xm cos(wt + ϕ) x (w)

a(t) = - w^2 xm cos(wt + ϕ) ⟶ Acceleration



Simple Harmonic Motion



The velocity and acceleration for simple harmonic motion can be found by differentiating the displacement:

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

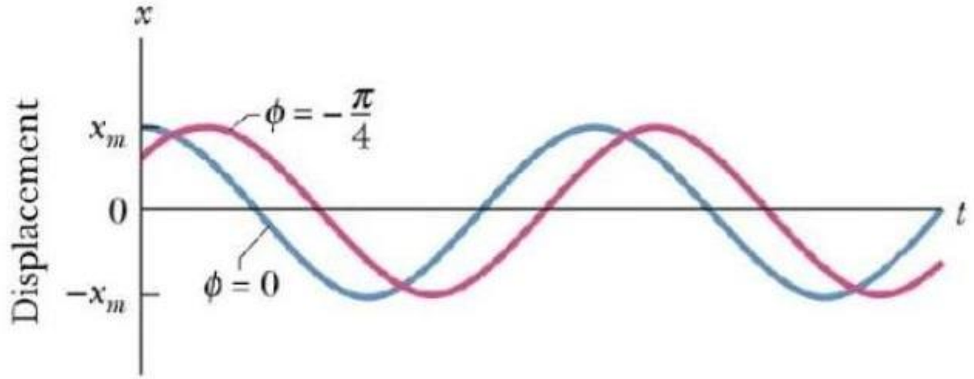
$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = -\omega^2 A \cos(\omega t + \phi).$$

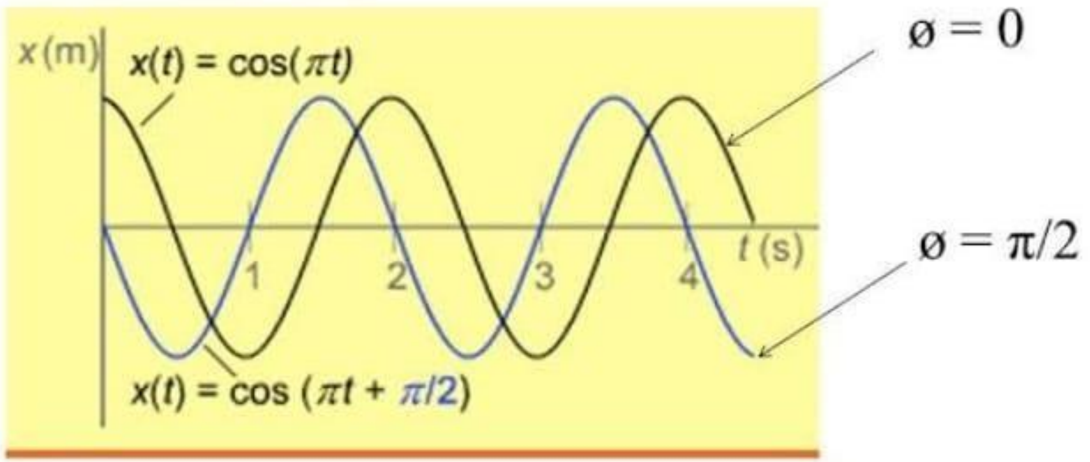
Phase and phase constant: $\boxed{x = x_m \cos(\omega t + ø)}$

Phase: the angle over which the function repeats itself-- $(\omega t + ø)$

Phase constant (ø) is the shift of the function if $x \neq x_m$ at $t = 0$ s

$$x(t) = A\cos(\omega t + \boxed{\phi})$$





Damped Harmonic Oscillation

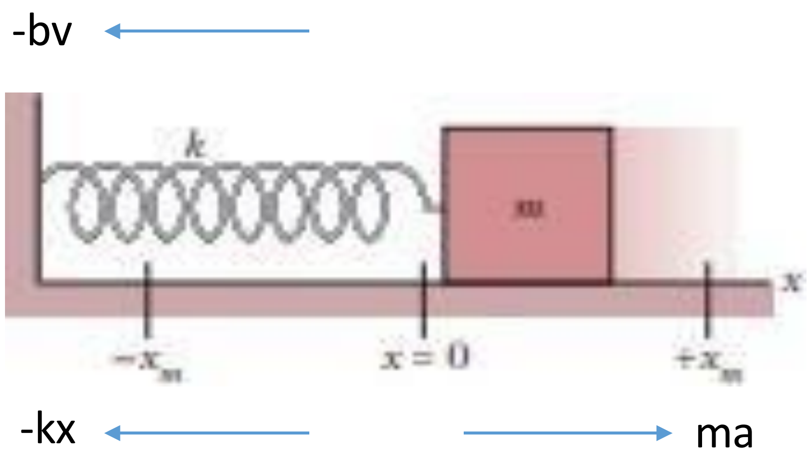
- Oscillation in the presence of frictional force
- When the motion of an oscillator is reduced by an external force, the oscillator and its motion is said to be damped

$$F_d \ \alpha \ v$$

$$F_d = -bv,$$

- "b" is the damping constant
- The net force along the x-axis will be:

$$-bv - kx = ma.$$



Damping

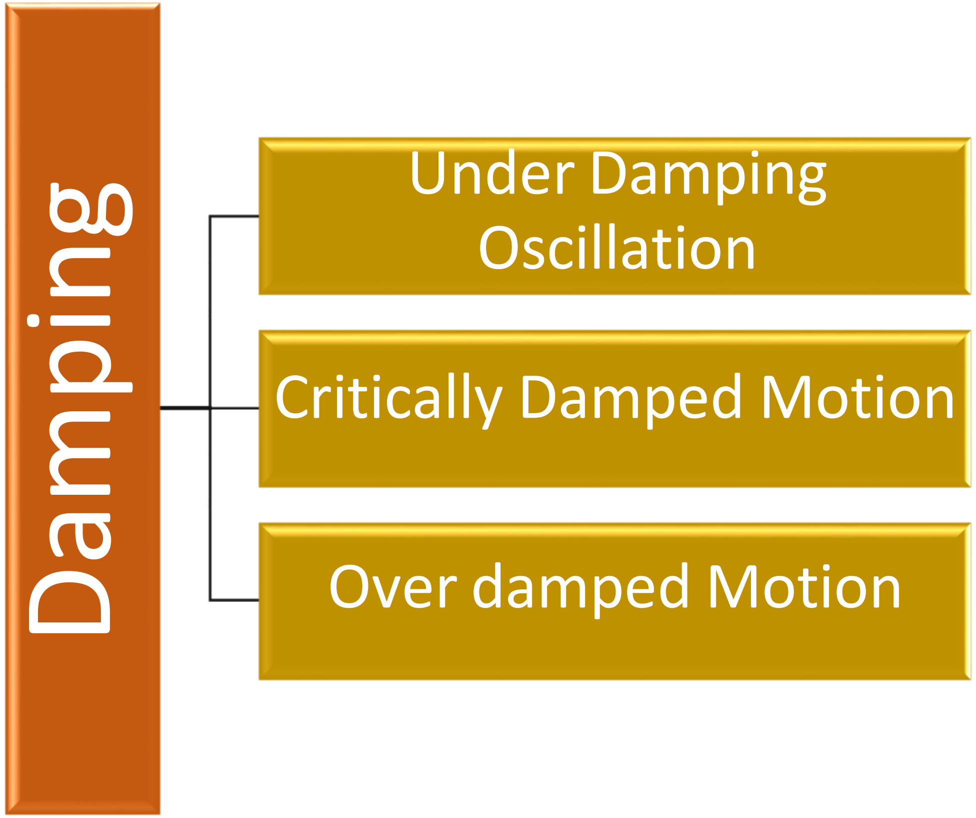
- All oscillators eventually stop oscillating as their amplitudes decrease rapidly, or gradually. This happens due to resistive forces, such friction or air resistance, which act in the opposite direction to the motion of an oscillator.

- Resistive forces acting on an oscillating simple harmonic system cause damping and it is known as damped harmonic oscillations.

- Damping is defined as:
  The reduction in energy and amplitude of oscillations due to resistive forces on the oscillating system

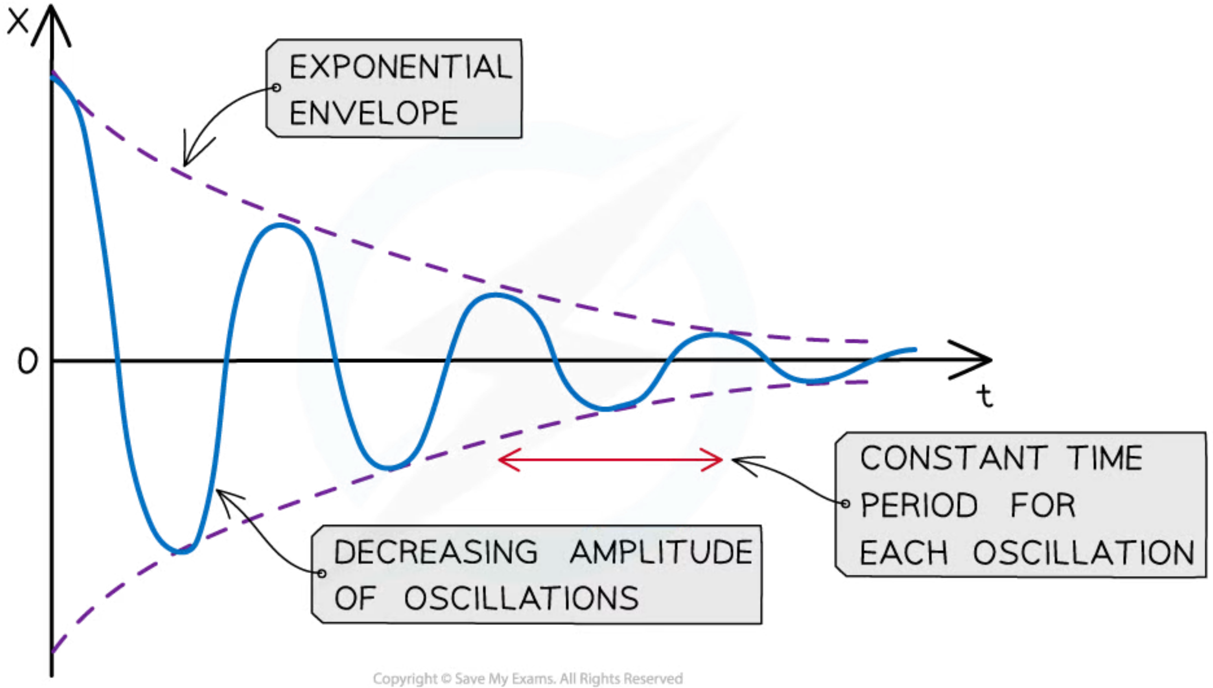
- Damping continues until the oscillator comes to rest at the equilibrium position

Types of Damping



Light Damping

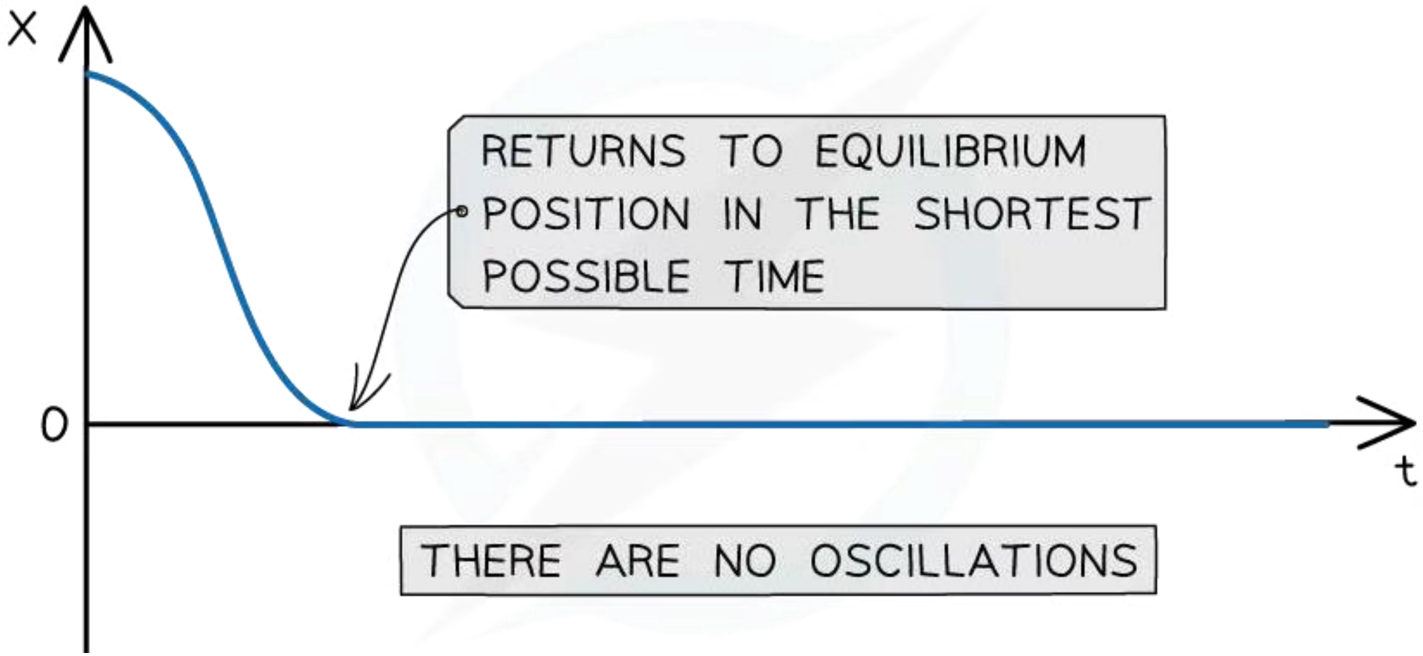
- When oscillations are lightly damped, the system will oscillate with gradually decreasing amplitude
- For example, a swinging pendulum decreasing in amplitude until it comes to a stop



Critical Damping

- When a critically damped oscillator is displaced from the equilibrium, it will return to rest at its equilibrium position in the shortest possible time without oscillating

- For example, car suspension systems prevent the car from oscillating after travelling over a bump in the road

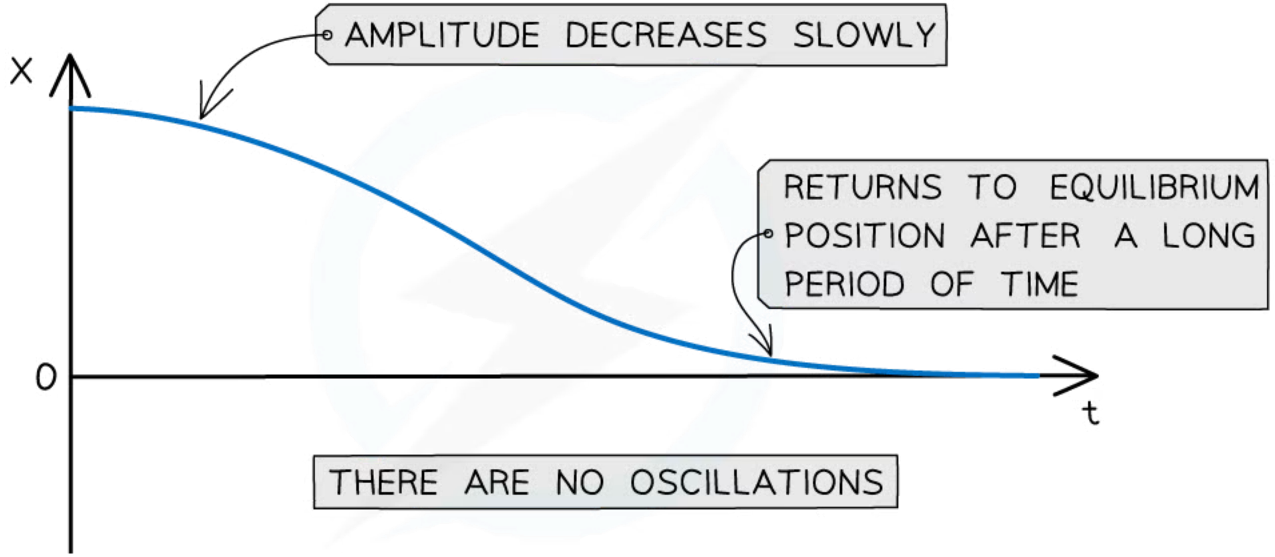


Heavy Damping

- When a heavily damped oscillator is displaced from the equilibrium, it will take a long time to return to its equilibrium position without oscillating

- The system returns to equilibrium more slowly than the critical damping case

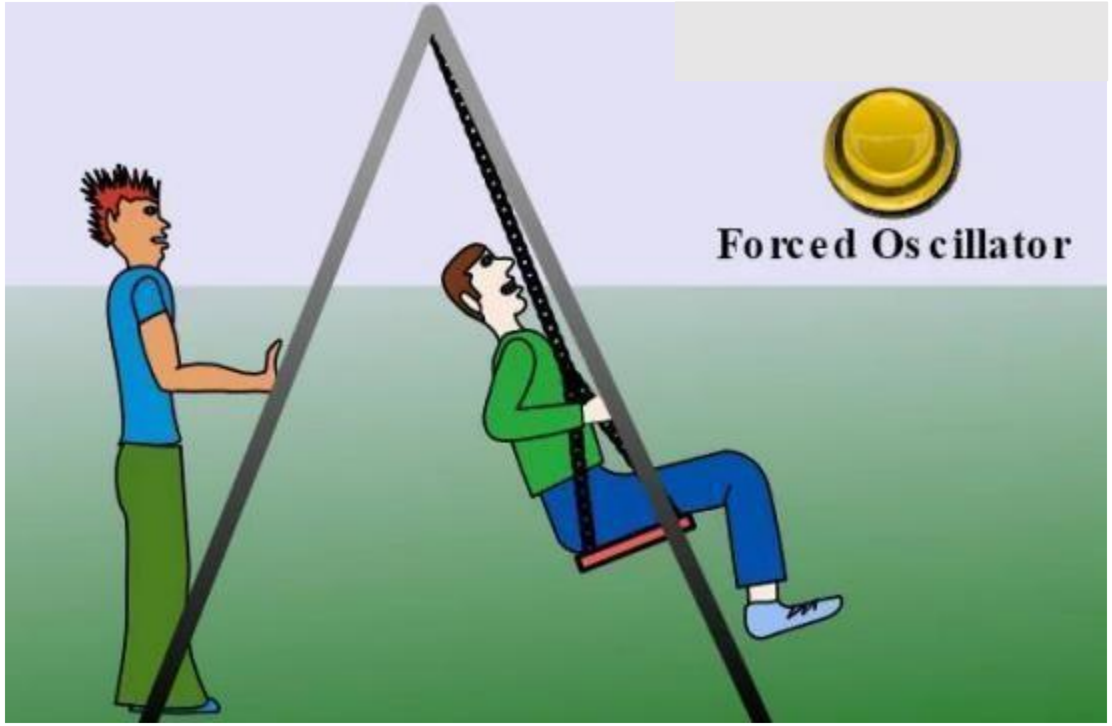
- For example, door dampers to prevent them slamming shut



heavy damping curve has no oscillations, and the displacement returns to zero after a long period of time

Force Oscillation

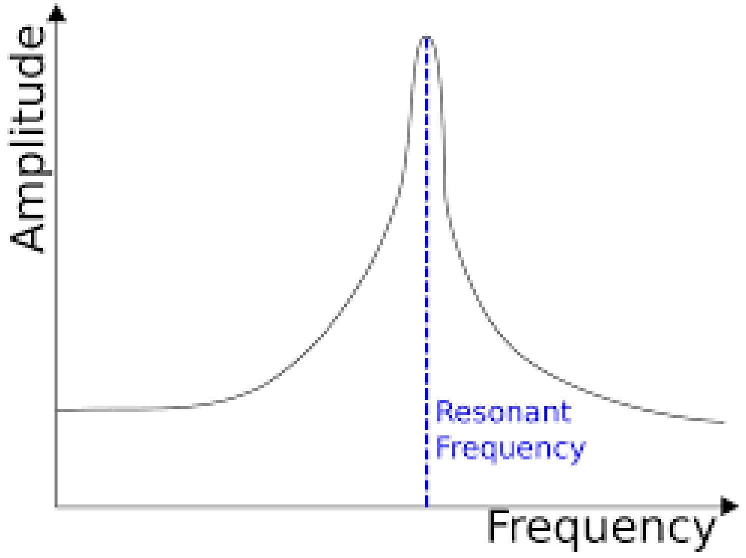
- Oscillation in the presence of an external force is called force oscillation



Resonance

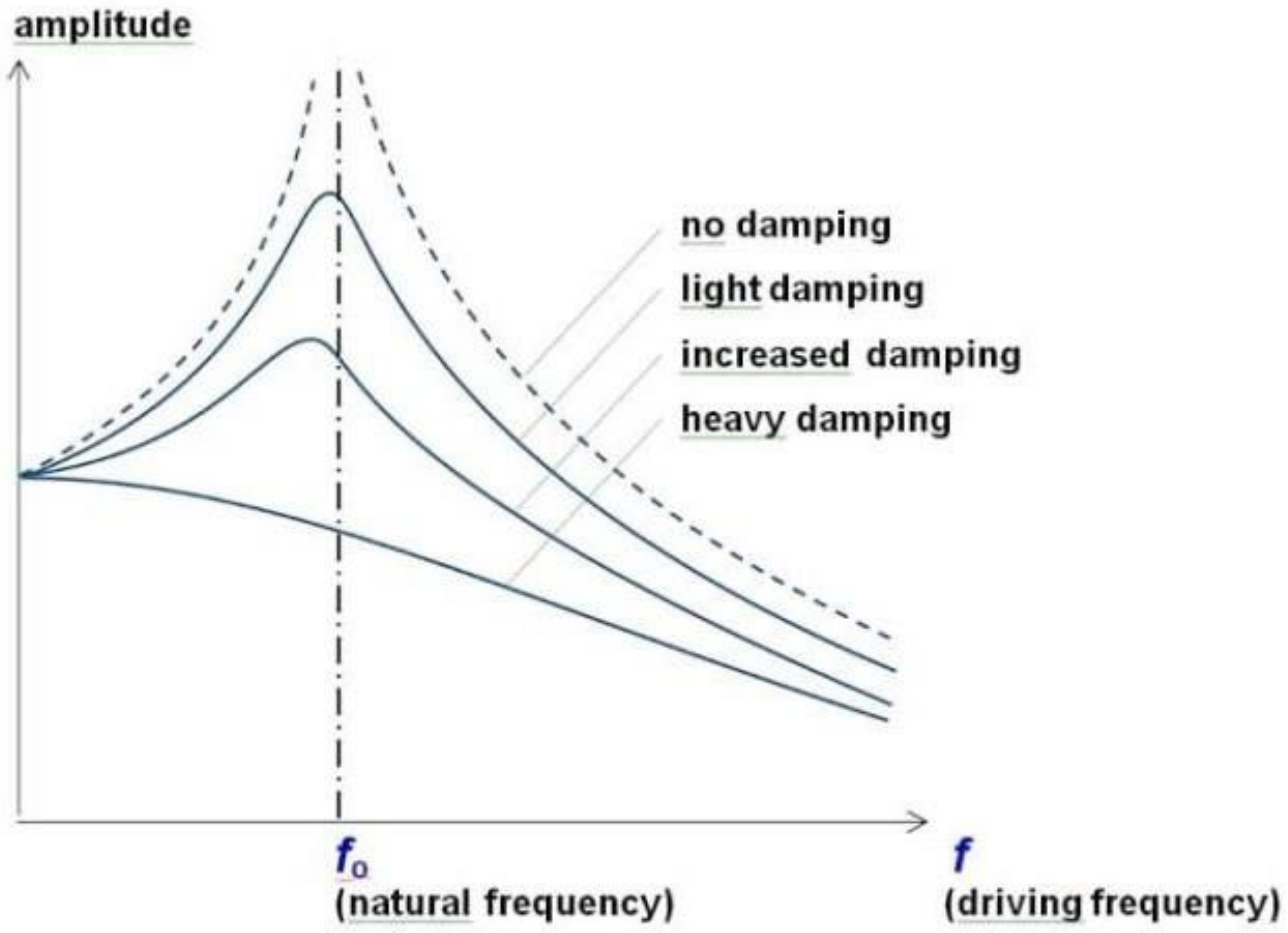
- Resonance occurs when an oscillating system is driven (made to oscillate from an outside source) at a frequency which is the same as its own natural frequency.

- Resonance occurs when the natural frequency “w” becomes close or equals to driven frequency “wd”.



Examples of Resonance

- Some opera singers are able to shatter glass by singing at its natural frequency.
- Sound produced by your vocal folds resonates in your larynx, throat, mouth, and nose, producing a much louder sound than the folds make on their own.
- Buildings and bridges have been known to resonate with earthquakes, wind, and pedestrians.


amplitude
no damping
light damping
increased damping
heavy damping
f_0
(natural frequency)
f
(driving frequency)